

# Population Distributions of Footbridge Modal Properties: A PRISMA-Guided Review and Hierarchical Bayesian Analysis

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## Abstract

*Keywords:*

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## 1. Introduction

The increasing adoption of light weight materials and slender structural forms has made modern footbridges more susceptible to human-induced vibrations. As a result, vibration serviceability under pedestrian loading has become an important design consideration to ensure acceptable structural performance and user comfort. Such assessment depends on several aspects of the bridge, including its structural form, material properties, and dynamic characteristics [1, 2, 3]. Therefore, prior knowledge or reliable estimation of key modal parameters is essential at the design stage, as these parameters provide an initial basis for assessing the feasibility of a proposed design [4, 5, 6, 7].

The current vibration serviceability codes and guidelines represent pedestrian loading through simplified load models to enable practical application at the design stage [8, 9, 10, 11, 12, 13, 14]. Since these load models are used to assess the vibration response of a wide range of footbridges, their adequacy should be evaluated in terms of the resulting structural response, rather than solely through how accurately they reproduce pedestrian loading.

The key limitation of many existing simplified load models is that pedestrians are represented as moving forces, with the corresponding load models calibrated without explicitly accounting for the dynamic interaction between the pedestrians and the supporting structure [15, 16, 17]. Human-structure interaction (HSI) alters the dynamic properties of the coupled pedestrian-structure system, and can lead to structural responses that differ substantially from those predicted by simplified load models. These effects can be

31 particularly important for lightweight structures and for cases with relatively  
32 large pedestrian-to-structure mass ratios [18, 17]. Consequently, simplified  
33 code models calibrated without accounting for HSI may exhibit systematic  
34 response bias that varies with the dynamic characteristics of the bridge, mo-  
35 tivating their reassessment and response-based recalibration.

36 Response-based calibration provides such a framework by comparing the  
37 response predicted by the simplified design model with that obtained from ex-  
38 perimental observations or a higher-fidelity reference model over a prescribed  
39 range of structural properties. Previous studies have adopted this approach  
40 by evaluating simplified pedestrian loading or vibration serviceability mod-  
41 els over ranges of natural frequency, damping and other structural param-  
42 eters [19, 3, 17, 18, 20]. However, these structural domains considered in  
43 such studies have generally been defined using prescribed parameter ranges,  
44 idealised structural configurations, or regularly sampled parameter combi-  
45 nations rather than using the observed statistical distribution of dynamic  
46 properties within the footbridge population. As a result, some structural  
47 conditions may receive disproportionate importance in the calibration sim-  
48 ply because of the adopted sampling scheme, rather than because they are  
49 common in practice.

50 For code-model calibration, it is important to know not only the range  
51 of structural properties that may occur, but also how common these proper-  
52 ties are within the footbridge population. Several recent studies have made  
53 valuable efforts to compile databases of footbridge characteristics and modal  
54 properties. Oliveira et al. (2014) [21] reported fundamental frequencies  
55 for 79 footbridges in Portugal, while Van Nimmen (2015) reviewed data  
56 for more than 160 footbridges [22, 23]. Wei et al. (2019) [24] assembled  
57 dynamic-property data for 138 footbridges, and Gong et al. (2024) [25]  
58 more recently studied a collection of footbridges in China. These studies  
59 provide a substantial starting point, but their datasets were assembled for  
60 different purposes, report different properties, and include some of the same  
61 bridges. Further observations are dispersed across many individual case stud-  
62 ies [26, 6, 27, 20]. Consolidating these records, identifying duplicate bridges,  
63 and recording which modal properties are available would provide a clearer  
64 basis for estimating their observed distributions.

65 This study addresses the above gap in two stages. First, a systematic  
66 literature search and screening procedure following the PRISMA framework  
67 is used to identify relevant publications and compile bridge-level data on key  
68 modal properties. Second, hierarchical Bayesian regression is used to proba-  
69 bilistically characterise incompletely observed modal properties while retain-  
70 ing the uncertainty associated with their estimation. The resulting database  
71 provides a population-level description of key footbridge modal properties

and their relationships, allowing the prevalence of different combinations of natural frequency, damping ratio and modal mass to be quantified. Such information provides a statistical basis for distinguishing commonly occurring structural conditions from those that are comparatively rare. This, in turn, can support the selection and weighting of representative structural parameter combinations used in vibration-serviceability assessment and code-model calibration.

Accordingly, this study aims to characterise the modal properties of the footbridge population using a systematically assembled bridge-level database of approximately 250 structures. The principal contributions of the study are: (i) the systematic identification and consolidation of published footbridge dynamic-property data; (ii) the construction of a large, non-duplicated database of bridge-level modal properties; (iii) the probabilistic treatment of incompletely observed damping and modal-mass information using hierarchical Bayesian regression; and (iv) the population-level characterisation of modal properties relevant to vibration-serviceability assessment and code-model calibration.

## 2. Systematic database development

The footbridge database was developed through a systematic literature identification and screening process following the PRISMA framework [28]. The procedure includes publication identification, title and abstract screening, full text eligibility assessment, bridge level data extraction and consolidation of information into a single database. The following subsections describe the search and screening procedure, the data extraction process, and the resulting database.

### 2.1. Review scope and search strategy

The systematic review aimed to identify published investigations reporting structural and dynamic properties of full scale pedestrian bridges. The principal variables of interest were bridge span, deck width, fundamental vertical natural frequency, vertical damping ratio, Modal mass, structural form, cross section, and construction materials.

Scopus was selected as the literature database because of its broad coverage of peer-reviewed engineering research [29]. The candidate publications were identified using Elsevier Scopus search API through title, abstract, and keyword fields. The search used a broader combination of pedestrian-bridge terminology with terms relating to natural frequency, damping, and structural mass, experimental dynamic testing, ambient vibration testing, operational modal analysis, model updating and vibration serviceability.

110 The search returned 727 bibliographic records before deduplication. Dub-  
111 plicate records were identified using DOI, Scopus EID. Following deduplica-  
112 tion, 697 unique records were retained for screening. A DOI was available for  
113 529 records, while Scopus EID were retained for publications without a DOI.  
114 Scopus records were used to identify and screen publications, while numerical  
115 bridge properties were extracted from the full texts of the eligible studies.

## 116 2.2. Eligibility criteria

117 Publications were considered eligible when they reported an investigation  
118 of an existing, full-scale pedestrian bridge and contained experimentally mea-  
119 sured or identified structural or modal properties that could be associated  
120 with a specific structure. Studies based only on laboratory specimens, scaled  
121 models, hypothetical structures, or purely numerical investigations were ex-  
122 cluded. Publications were also excluded where the reported frequency could  
123 not be identified as a structural vertical natural frequency relevant to the  
124 present study. For inclusion in the bridge-level database, sufficient informa-  
125 tion was required to identify the bridge and extract its fundamental vertical  
126 natural frequency together with the geometric information required for the  
127 population analysis. Damping ratio, modal mass, structural form, cross-  
128 section, and construction material were extracted where available but were  
129 not mandatory for inclusion.

## 130 2.3. Study screening and bridge identification

131 Following bibliographic deduplication, the 697 unique publications were  
132 retained for full-text screening. The full-text documents were processed  
133 using a semi-automated Python workflow. PDF text was extracted using  
134 PyMuPDF and subsequently searched for evidence of full-scale pedestrian  
135 bridge investigations and relevant structural and modal properties.

136 The extracted text was searched for bridge names and terminology as-  
137 sociated with natural frequency, damping, modal mass, mode shapes, and  
138 experimental vibration testing. Reference sections were excluded from the  
139 search to reduce the likelihood of identifying bridges mentioned only in cited  
140 literature.

141 During the screening stage, each publication was counted as a single  
142 record, regardless of how many bridges it investigated. During data ex-  
143 traction, however, each individual bridge was treated as a separate database  
144 record. Consequently, a single publication could contribute data for several  
145 bridges.

146 Each detected bridge was assigned a separate candidate record containing  
147 its source publication, bridge identity, and associated structural and modal-  
148 property evidence. Candidate records were subsequently reviewed against

the bridge-level inclusion requirements defined in Section 2.2 before consolidation.

#### *2.4. Bridge-level data extraction and consolidation*

For each candidate bridge, the extracted information included bridge name and location, span, deck width, fundamental vertical natural frequency, vertical damping ratio, modal mass, structural form, cross-section, and construction material. Numerical quantities were converted to consistent units before consolidation.

Because the same physical bridge could be reported in several publications, repeated bridge records were identified using bridge name, location, geometry, structural characteristics, reported modal properties, and source references. Potential matches across different publications were reviewed manually rather than being automatically removed.

Where several publications referred to the same physical bridge, the available information was consolidated into a single bridge-level record. The original experimental investigation was retained as the principal source where available, while additional information from later publications was incorporated where appropriate.

#### *2.5. Resulting database and data completeness*

Following bridge-level screening and duplicate reconciliation, 250 unique footbridges were retained in the final database. Fundamental vertical natural frequency and span were available for all bridges, while deck width was available for 240 bridges. Vertical damping ratios were reported for 100 bridges, whereas mass related information was considerably less complete, with modal mass information available for only 30 bridges and linear mass available for 25 bridges. Structural form and material information were available for all bridges, while cross-section information was available for 242 bridges. The resulting differences in data completeness, particularly for damping and mass-related properties, motivated the probabilistic treatment introduced in the following section.

### **3. Bayesian treatment of incomplete bridge data**

The database contains complete records of fundamental vertical natural frequency and span, but damping and mass-related properties are reported for only a subset of bridges. Analysing only bridges with complete records would therefore exclude a substantial proportion of the database. To retain these bridges in the population analysis, Bayesian regression models were

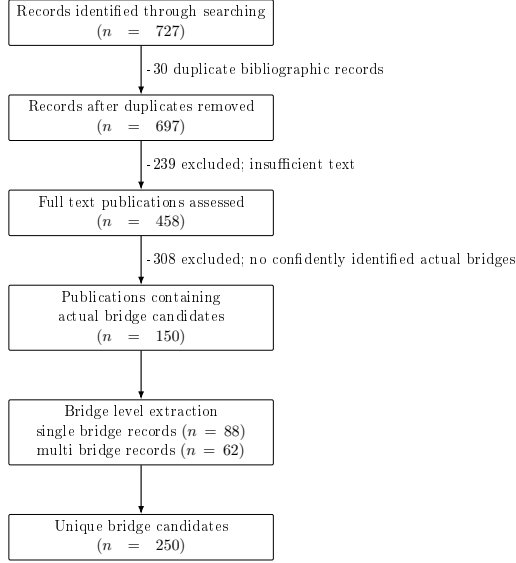


Figure 1: PRISMA flow diagram for the systematic identification, screening, and consolidation of footbridge data.

185 developed to infer the unreported properties from the available bridge char-  
 186 acteristics. The missing values were represented by posterior predictive dis-  
 187 tributions, allowing uncertainty in both regression parameters and remaining  
 188 variability between bridges to be propagated into resulting population dis-  
 189 tributions.

### 190 3.1. Hierarchical model for vertical damping ratio

191 Let  $\xi_i$  denote the damping ratio of the first vertical mode for the  $i$ -th  
 192 bridge. The damping ratios  $\xi_i$  differ between bridges, reflecting differences  
 193 in their construction materials, structural forms, and other factors. How-  
 194 ever, the database does not provide sufficient details to model the physical  
 195 mechanisms governing damping in each bridge. Statistical modelling offers a  
 196 means of estimating unreported values from observed data without explicitly  
 197 identifying the underlying mechanisms [30, 31, 32]. Accordingly, a Bayesian  
 198 hierarchical regression model was fitted to the reported damping ratios, us-  
 199 ing natural frequency, span, and material as predictors that are available in  
 200 the database. The fitted model was then used to obtain posterior predictive  
 201 distributions for unreported damping ratios, accounting for both uncertainty  
 202 in the regression parameters and remaining variability between bridges. The  
 203 probabilistic structure of the hierarchical damping model is shown in Fig. 2.

204  
 205 For each bridge, the available predictor information is  $x_i = \{f_i, L_i, m_i\}$ ,

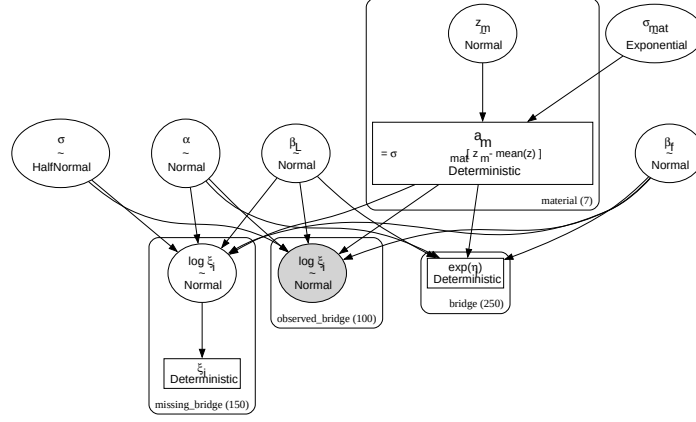


Figure 2: Probabilistic structure of the hierarchical Bayesian model for vertical damping ratio.

where  $f_i$  is the fundamental vertical natural frequency,  $L_i$  is the span, and  $m_i$  denotes the material category. Let  $\mathcal{O}$  and  $\mathcal{M}$  denote the sets of bridges with observed and missing damping ratios, respectively. The data used to estimate the model parameters are then

$$\mathcal{D}_{\text{obs}} = \{(\xi_i, \mathbf{x}_i) : i \in \mathcal{O}\}. \quad (1)$$

Damping ratios were modelled using a lognormal distribution to ensure positive predictions, with the corresponding log damping ratios assumed to follow a normal distribution:

$$\log \xi_i \mid \eta_i, \sigma \sim \mathcal{N}(\eta_i, \sigma^2), \quad (2)$$

where  $\eta_i$  is the conditional mean of log damping ratio and  $\sigma$  describes the remaining variability between bridges. The conditional mean was expressed as

$$\eta_i = \alpha + \beta_f x_{f,i} + \beta_L x_{L,i} + a_{m_i}, \quad (3)$$

where  $\alpha$  is the common intercept,  $\beta_f$  and  $\beta_L$  describe the associations with natural frequency and span, and  $a_{m_i}$  is the effect of the material category of bridge  $i$ . The predictors  $f_i$  and  $L_i$  were logarithmically transformed and standardised as,

$$x_{f,i} = \frac{\log f_i - \overline{\log f}}{s_{\log f}}, \quad x_{L,i} = \frac{\log L_i - \overline{\log L}}{s_{\log L}}. \quad (4)$$

Here, the overbars and  $s$  denote the means and standard deviations of the corresponding log-transformed predictors.

222 The material categories contain different numbers of observed damping  
 223 ratios. Estimating an independent model for each category would therefore  
 224 provide limited information for sparsely represented materials. Instead, ma-  
 225 terial effects were assigned a shared distribution,

$$a_m \mid \sigma_{\text{mat}} \sim \mathcal{N}(0, \sigma_{\text{mat}}^2). \quad (5)$$

226 This hierarchical structure allows information to be partially pooled across  
 227 materials while retaining differences between categories. The parameter  $\sigma_{\text{mat}}$   
 228 controls the variation in material effects around the common baseline.

229 Let  $\boldsymbol{\theta}$  contain the regression coefficients, material effects, and dispersion  
 230 parameters:

$$\boldsymbol{\theta} = \{\alpha, \beta_f, \beta_L, \mathbf{a}, \sigma, \sigma_{\text{mat}}\}. \quad (6)$$

231 Assuming conditional independence between bridge records, the likelihood  
 232 of the observed log damping ratios is

$$\mathcal{L}(\mathcal{D}_{\text{obs}} \mid \boldsymbol{\theta}) = \prod_{i \in \mathcal{O}} \mathcal{N}(\log \xi_i \mid \eta_i, \sigma^2). \quad (7)$$

233 The posterior distribution follows from Bayes' rule:

$$p(\boldsymbol{\theta} \mid \mathcal{D}_{\text{obs}}) \propto \mathcal{L}(\mathcal{D}_{\text{obs}} \mid \boldsymbol{\theta}) p(\boldsymbol{\theta}). \quad (8)$$

234 For a bridge with unreported damping, its known predictors were used to  
 235 obtain the posterior predictive distribution:

$$p(\xi_i \mid \mathbf{x}_i, \mathcal{D}_{\text{obs}}) = \int p(\xi_i \mid \mathbf{x}_i, \boldsymbol{\theta}) p(\boldsymbol{\theta} \mid \mathcal{D}_{\text{obs}}) d\boldsymbol{\theta}, \quad i \in \mathcal{M}. \quad (9)$$

236 This distribution accounts for both uncertainty in the model parameters and  
 237 the remaining variability between bridges. Samples from these distributions  
 238 were used to represent the missing damping ratios in the subsequent popu-  
 239 lation analysis.

### 240 3.2. Bayesian estimation of pedestrian-to-structure mass-ratio probabilities

241 The pedestrian-to-structure mass ratio depends on both the bridge modal  
 242 properties and the distribution of pedestrians. The present analysis therefore  
 243 considers the prevalence of mass ratios at a specified pedestrian density  $\rho$ .

244 For bridge  $i$ , assuming pedestrians are uniformly distributed over the  
 245 occupied deck area  $A_i$ , the pedestrian modal mass associated with the first  
 246 vertical mode is

$$M_{p,i}^*(\rho) = \rho \bar{m}_p \int_{A_i} \phi_i^2(\mathbf{x}) dA, \quad (10)$$

247 where  $\bar{m}_p$  is the assumed mean pedestrian mass and  $\phi_i$  is the corresponding  
 248 mode shape. The pedestrian-to-structure modal mass ratio is then,

$$\mu_i(\rho) = \frac{M_{p,i}^*(\rho)}{M_{b,i}^*}, \quad (11)$$

249 where  $M_{b,i}^*$  is the bridge modal mass evaluated using the same mode-shape  
 250  $\phi_i$ .

251 Sufficient information to calculate modal mass ratios is available for only  
 252 a subset of bridges in the database. A Bayesian model was therefore used  
 253 to estimate the proportion of bridges whose mass ratios fall below a spec-  
 254 ified threshold  $T$ , without estimating a missing mass ratio for each bridge.  
 255 Threshold probabilities were estimated separately for each material category  
 256 and then combined according to the estimated prevalence of these categories.  
 257 The probabilistic structure of the Bayesian mass-ratio model is shown in Fig-  
 ure 3.

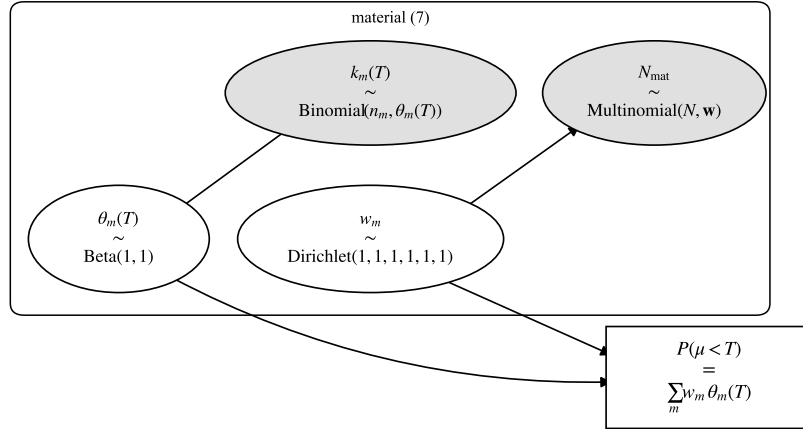


Figure 3: Probabilistic structure of the Bayesian model for pedestrian-to-structure mass-ratio probabilities.

258 For a material category  $m$ , let  $\theta_m(T; \rho)$  be,  
 259

$$\theta_m(T; \rho) = P(\mu(\rho) < T \mid m). \quad (12)$$

260 Let  $n_m$  be the number of bridges with calculable mass ratios in category  $m$ ,  
 261 and let  $k_m(T; \rho)$  be the number of these bridges whose mass ratios are below  
 262 the threshold  $T$ . The first likelihood related these counts to the material  
 263 specific threshold probability:

$$k_m(T; \rho) \sim \text{Binomial}(n_m, \theta_m(T; \rho)), \quad (13)$$

264 This formulation assumes that bridges with calculable mass ratios are repre-  
 265 sentative of their material category with respect to mass ratio.

266 The second likelihood describes the material composition of the full  
 267 database. Let  $N_m$  be the total number of bridges in material category  $m$ ,  
 268 including those without calculable mass ratios, and let  $w_m = P(m)$  denote  
 269 the corresponding population proportion. For  $M$  mutually exclusive and  
 270 exhaustive material categories,

$$N_{mat} \mid w \sim \text{Multinomial}(N, w), \quad (14)$$

271 where  $N_{mat} = (N_1, \dots, N_M)$ ,  $w = (w_1, \dots, w_M)$ ,  $N = \sum_{m=1}^M N_m$ , and  $\sum_{m=1}^M w_m =$   
 272 1. Consequently, all bridges contribute to estimating the material propor-  
 273 tions, while only bridges with calculable mass ratios contribute to estimating  
 274 the material-specific threshold probabilities. The prior distributions for these  
 275 parameters are specified in the following subsection.

276 The material proportions and threshold probabilities were combined using  
 277 the law of total probability:

$$\begin{aligned} P(\mu(\rho) < T) &= \sum_{m=1}^M P(m) P(\mu(\rho) < T \mid m) \\ &= \sum_{m=1}^M w_m \theta_m(T; \rho). \end{aligned} \quad (15)$$

278 Each product gives the probability that a bridge belongs to material category  
 279  $m$  and has a mass ratio below  $T$ . Summing across the material categories  
 280 give the overall proportion of bridges below the threshold.

281 To propagate uncertainty in both components, the overall probability was  
 282 evaluated for each posterior sample  $s$ :

$$F_{\mu}^{(s)}(T; \rho) = \sum_{m=1}^M w_m^{(s)} \theta_m^{(s)}(T; \rho). \quad (16)$$

283 The resulting samples account for uncertainty in both the material propor-  
 284 tions and the material-specific threshold probabilities.

285 The calculation was repeated over a range of thresholds  $T$  at a specified  
 286 pedestrian density  $\rho$  to characterise the cumulative distribution of pedestrian-  
 287 to-structure mass ratio. At each threshold, the posterior mean and 95%  
 288 credible interval were calculated for  $P(\mu(\rho) < T)$ . Lower and upper bound  
 289 of these pointwise credible intervals define the bounding curves of a proba-  
 290 bility box (p-box), allowing the estimated proportion of bridges below any  
 291 selected mass ratio threshold to be assessed together with its uncertainty.

### 292 3.3. Prior specification and posterior computation

293 As described in the Section 3.1, the damping model contains 12 unknown  
 294 model parameters, including the intercept  $\alpha$ , two regression coefficients  $\beta_f$   
 295 and  $\beta_L$ , the residual standard deviation  $\sigma$ , and the between-material standard  
 296 deviation  $\sigma_{\text{mat}}$ . The remaining seven parameters are the material-specific ef-  
 297 fects  $a_m$ , corresponding to the seven material categories represented in the  
 298 database: steel, concrete, composite, fibre-reinforced polymer (FRP), alu-  
 299 minium, timber, and other materials. Unreported damping ratios are addi-  
 300 tional latent quantities whose distributions are inferred from the model.

301 The intercept and regression coefficients were assigned normal priors:

$$\alpha \sim \mathcal{N}(\log(0.01), 0.5^2), \quad \beta_f, \beta_L \sim \mathcal{N}(0, 0.25^2), \quad (17)$$

302 where the second argument denotes the variance. The prior for  $\alpha$  was cen-  
 303 tred at  $\log(0.01)$ , corresponding to a baseline damping ratio of 1%. This  
 304 value was selected to place the prior within the range commonly associated  
 305 with footbridge vibration serviceability, where damping ratios are generally  
 306 between 0.4% to 2% [9, 13, 14] for vibrations under pedestrian excitaions  
 307 and vary with construction material. The relatively broad prior on the loga-  
 308 rithmic scale allows the observations and material specific effects to shift the  
 309 baseline towards both lower and higher values.

310 The regression coefficients were assigned zero-mean priors, reflecting the  
 311 absence of prior knowledge about the direction of their effects and to avoid  
 312 favouring any particular relationship between damping and the predictors.  
 313 Because the log-transformed predictors were standardised, the same prior  
 314 scale was used for both coefficients, regularising large effects while allowing  
 315 the observations to inform their direction and magnitude.

316 The material-specific effects were assigned a shared normal prior,

$$a_m \mid \sigma_{\text{mat}} \sim \mathcal{N}(0, \sigma_{\text{mat}}^2). \quad (18)$$

317 This prior treats the material effects as deviations from the common re-  
 318 gression relationship, centered at zero. Sharing the same distribution across  
 319 material categories allows partial pooling, so that material categories with  
 320 few observations can borrow information from those with more data, while  
 321 still allowing for differences between categories. The between-material stan-  
 322 dard deviation controls the extent of variation among the material effects  
 323 and was assigned the prior

$$\sigma_{\text{mat}} \sim \text{Exponential}(6). \quad (19)$$

324 An exponential prior was used because  $\sigma_{\text{mat}}$  is a positive scale parameter.  
 325 This prior restricts  $\sigma_{\text{mat}}$  to positive values and places greater prior probability

on relatively small differences between material categories while retaining support for larger differences if required by the data.

The remaining bridge-to-bridge variability not explained by the predictors or material effects is represented by the residual standard deviation  $\sigma$ , for which the prior

$$\sigma \sim \text{HalfNormal}(0.3) \quad (20)$$

was adopted. The half-normal prior similarly restricts the residual standard deviation to positive values while allowing substantial unexplained variability between bridges.

For computation, the material effects were expressed using a non-centred parameterisation with a sum-to-zero constraint,

$$z_m \sim \mathcal{N}(0, 1), \quad \bar{z} = \frac{1}{M} \sum_{j=1}^M z_j, \quad a_m = \sigma_{\text{mat}} (z_m - \bar{z}). \quad (21)$$

The centring ensures that  $\sum_{m=1}^M a_m = 0$ , so that the material effects represent deviations from the common intercept  $\alpha$ .

For the mass-ratio model described in Section 3.2, prior distributions were required for the material proportions  $w_m$  and the material-specific threshold probabilities  $\theta_m(T; \rho)$ . The material proportions were assigned a symmetric Dirichlet prior,

$$w \sim \text{Dirichlet}(1, \dots, 1), \quad (22)$$

where the number of concentration parameters corresponds to the number of material categories considered in the mass-ratio analysis. This prior treats the material categories symmetrically and does not favour any particular composition of the bridge population before observing the material counts in the database.

The material-specific threshold probabilities were assigned independent uniform beta priors,

$$\theta_m(T; \rho) \sim \text{Beta}(1, 1). \quad (23)$$

This prior assigns equal density to all probabilities between zero and one and therefore does not favour a particular probability that the mass ratio falls below the selected threshold within any material category. The same prior specification for  $\theta_m(T; \rho)$  was applied independently at each threshold  $T$ , whereas the material proportions  $w$  describe the common material composition of the database and are therefore unchanged across thresholds.

Both Bayesian models were implemented in PyMC and posterior sampling was performed using the No-U-Turn Sampler (NUTS), an adaptive Hamiltonian Monte Carlo algorithm. For each model, four independent chains were

run with 2000 tuning iterations followed by 2000 retained posterior samples per chain, using a target acceptance probability of 95%.

### 3.4. Model assessment procedures

#### 3.4.1. Vertical damping model

The vertical damping model was assessed through prior predictive checks, sampling diagnostics, comparison of prior and posterior parameter distributions, and posterior predictive checks. Prior predictive sampling was first used to examine whether the specified priors produced plausible damping ratios over the range of bridge characteristics represented in the database. Figure 4 compares the resulting prior predictive PDF and CDF with the observed damping ratios. As shown in Figure 4(a), the prior predictive distribution is concentrated in the low-damping region, where most of the observations occur, while retaining support over the upper range of the observed damping ratios. This is also evident from Figure 4(b), where the observed empirical CDF remains within the 95% prior predictive interval over the considered range. No substantial prior–data conflict was identified. The adopted priors were therefore considered sufficiently broad to represent plausible damping behaviour without placing excessive probability on unrealistically large damping ratios.

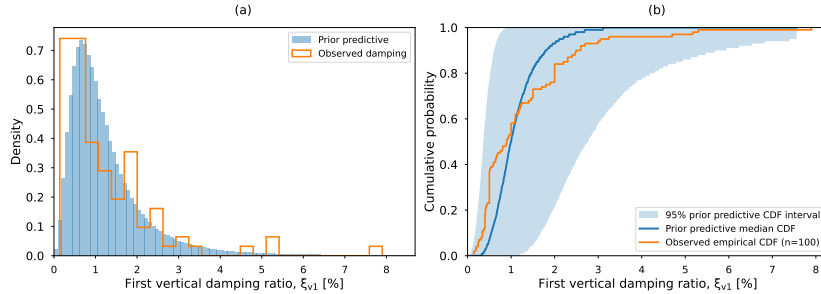


Figure 4: Prior predictive assessment of the vertical damping model: (a) prior predictive and observed damping distributions; and (b) prior predictive median CDF and 95% interval compared with the observed empirical CDF.

Posterior sampling performance was assessed using the marginal posterior distributions and trace plots shown in Figure 5. The four chains exhibited substantial overlap for the global parameters and material-specific effects, while the trace histories showed stationary behaviour with no persistent trends or visible separation between chains. No divergent transitions were observed. The potential scale reduction statistics were close to unity for all monitored parameters ( $\hat{R} = 0.9999$ – $1.0010$ ), with minimum bulk and tail effective sample sizes of approximately 2505 and 2294, respectively. These

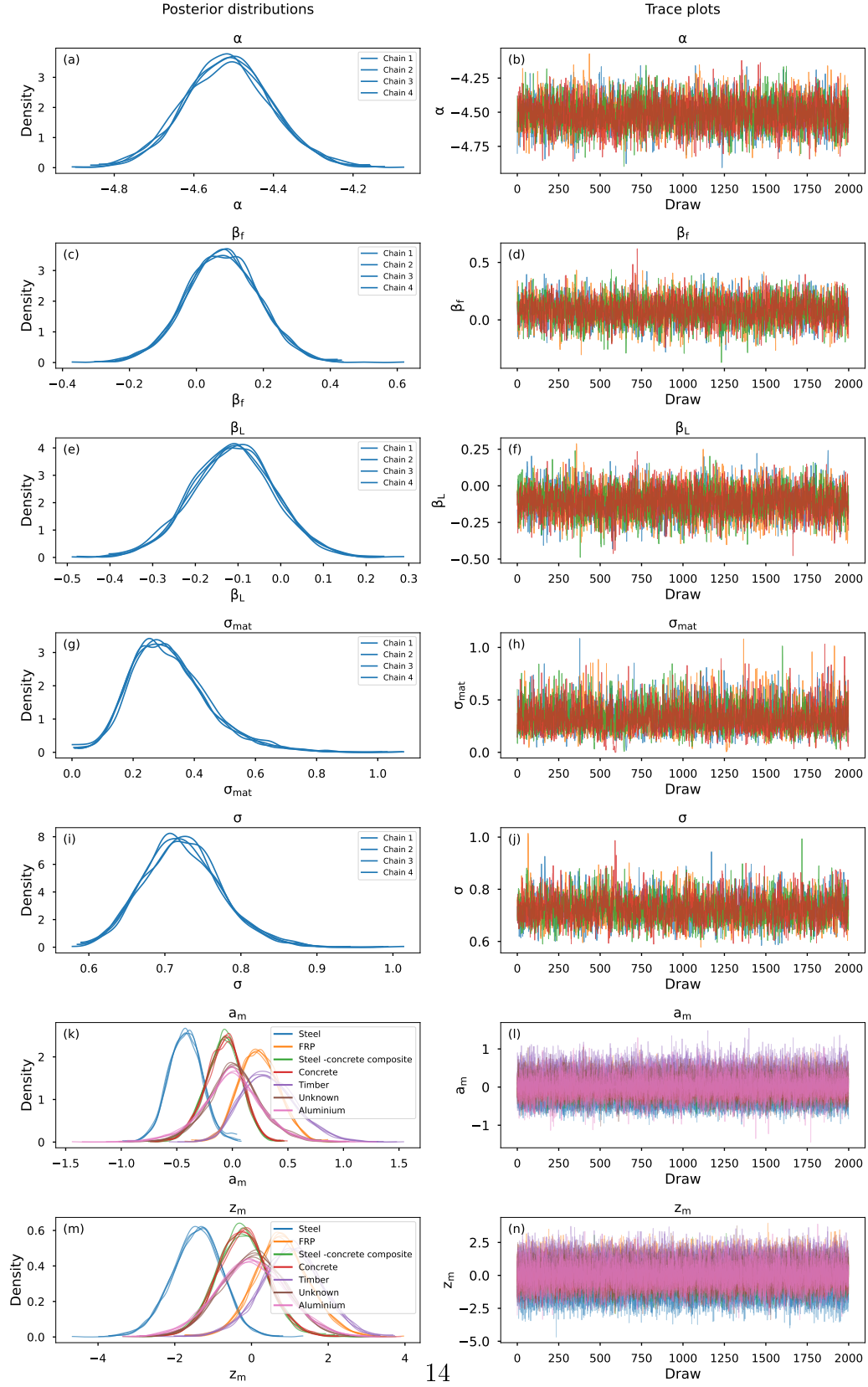


Figure 5: Posterior sampling diagnostics for the vertical damping model. The left column shows marginal posterior distributions for the four sampling chains and the right column shows the corresponding trace plots for the global parameters and material-specific effects.

385 diagnostics indicated satisfactory convergence and effective exploration of the  
386 posterior distribution.

387 The extent to which the observed damping data informed the model pa-  
388 rameters was examined by comparing the prior and posterior distributions  
389 in Figure 6. The posterior distributions are generally more concentrated  
390 than their corresponding priors, indicating that the observed data provide  
391 substantial information for the model parameters.

392 The posterior distribution of the intercept  $\alpha$  is considerably narrower than  
393 its prior and is centred at approximately  $-4.51$ , corresponding to a condi-  
394 tional median damping ratio of approximately  $1.1\%$  at the reference values of  
395 the standardised predictors and zero material effect. The posterior distribu-  
396 tions of  $\beta_f$  and  $\beta_L$  indicates a tendency for damping to increase with natural  
397 frequency and decrease with span respectively. However, both posterior dis-  
398 tributions retain an appreciable probability on either side of zero, indicating  
399 uncertainty in the strength and direction of these effects.

400 The posterior distribution of  $\sigma_{\text{mat}}$  is shifted away from the strong con-  
401 centration near zero imposed by its prior, indicating that the data support  
402 variability in damping between material categories. Similarly, the posterior  
403 distribution of the residual standard deviation  $\sigma$  is strongly concentrated rel-  
404 ative to its prior, indicating substantial remaining bridge-to-bridge variability  
405 after accounting for frequency, span, and material.

406 The material-specific posterior distributions have different drifts from-  
407 common baseline. Steel exhibits a clear negative material effect, indicating  
408 lower damping relative to the overall baseline. FRP and timber show ten-  
409 dencies towards positive material effects, whereas steel-concrete composite  
410 and concrete show slight negative material. The posterior distributions for  
411 the unknown-material and aluminium categories remain relatively broad and  
412 centred close to zero, reflecting the limited information available for these  
413 two categories.

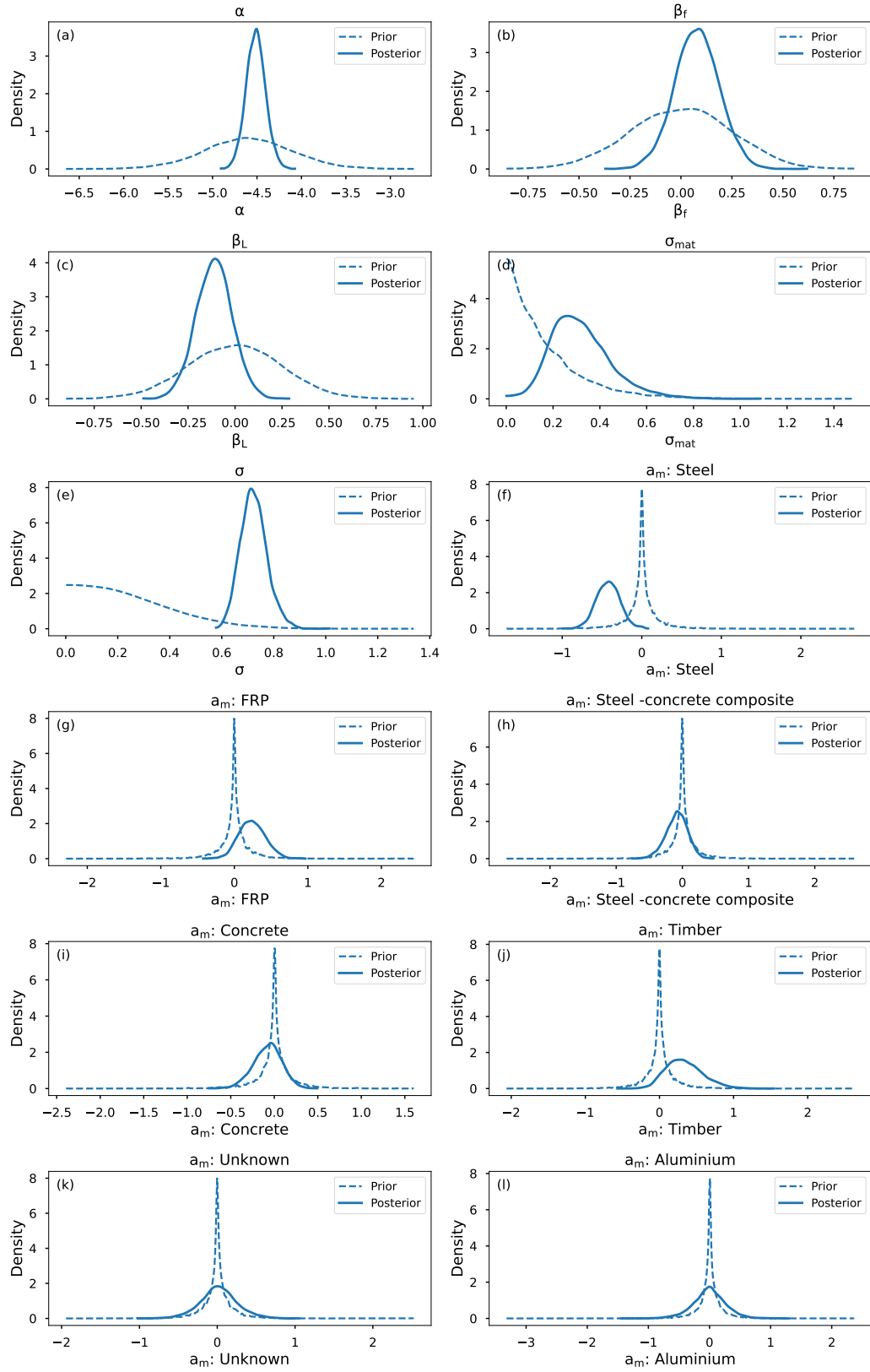


Figure 6: Comparison of prior and posterior distributions.

414 3.4.2. *Mass-ratio probability model*

## 415 4. Results

416 4.1. *Database characteristics and observed distributions*

417 4.2. *Population distribution of vertical damping ratio*

418 4.3. *Population distribution of pedestrian-to-structure mass ratio*

419 4.4. *Joint prevalence of serviceability-relevant properties*

## 420 5. Discussion

## 421 6. Conclusions

## 422 7. Acknowledgements

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425 Scholarship (MITS).

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